CSI Data Preprocessing Final Report

1. Pre-Processing Parameters

Stage	Parameters / Methodology
Balancing	Uniform length: 13,590 packets
Subcarrier Removal	Indices: 0-5, 11, 25, 32, 39, 53, 58-63
Hampel Filter	Window=5, Threshold=3-Sigma
Butterworth Filter	4th Order, 10Hz Cutoff, 100Hz Sampling
Dimension Reduction	PCA: top 10 principal components
Normalization	Min-Max scaling to [0.0, 1.0]

2. Visual Verification (Raw vs. Preprocessed)

1. We have collected data for 3 subjects.
   5 Activities (Walk, Run, Jump, Sit/Stand and Fall)
2. First, we visualized the data without any pre-processing (i.e. Cleaning, Balancing)
3. By applying different pre-processing techniques, we again visualize the data and check whether our data collection is correct or not.
4. At last, we compared our data with different research papers and get verification that our data is collected correctly.
5. We visualized the data for 1 sec, 10 sec and 1 min but we will see visualization of data for only 1 sec and will discuss further in meeting.

Here is the simple visualization comparison of raw and pre-processed data:

1. Subject 1

1. Walk

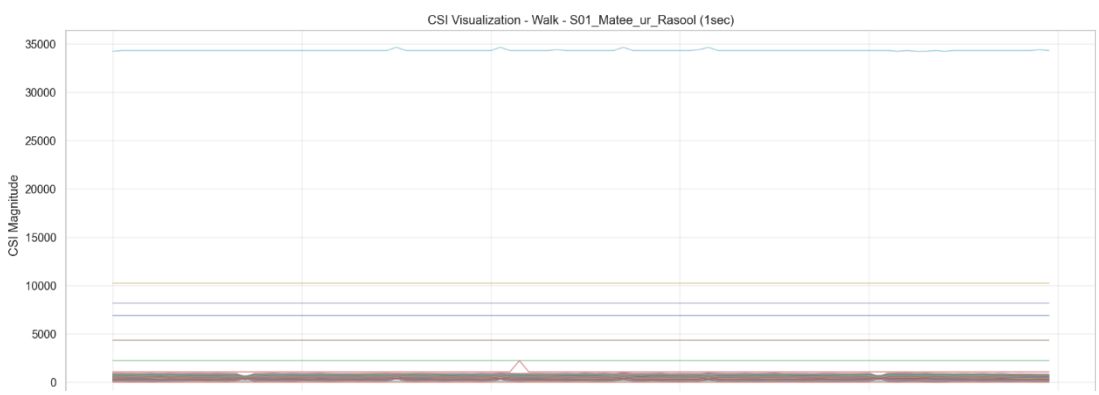


Figure 1.Raw data

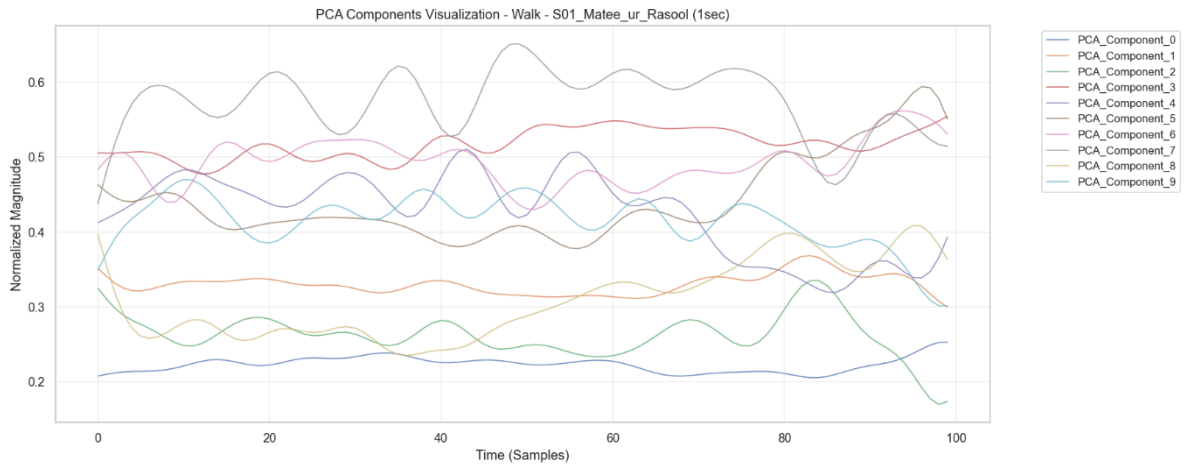


Figure 2: Processed Data

2. Run

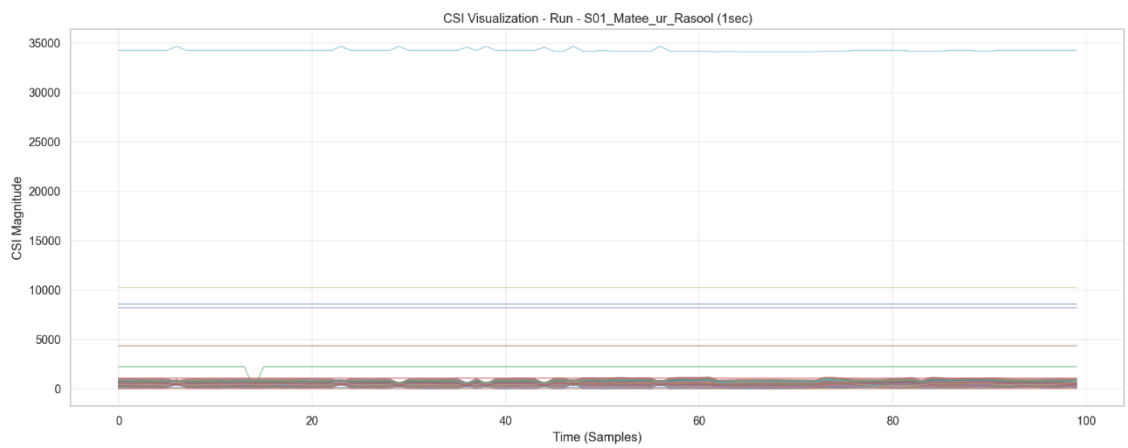


Figure 4:Raw Data

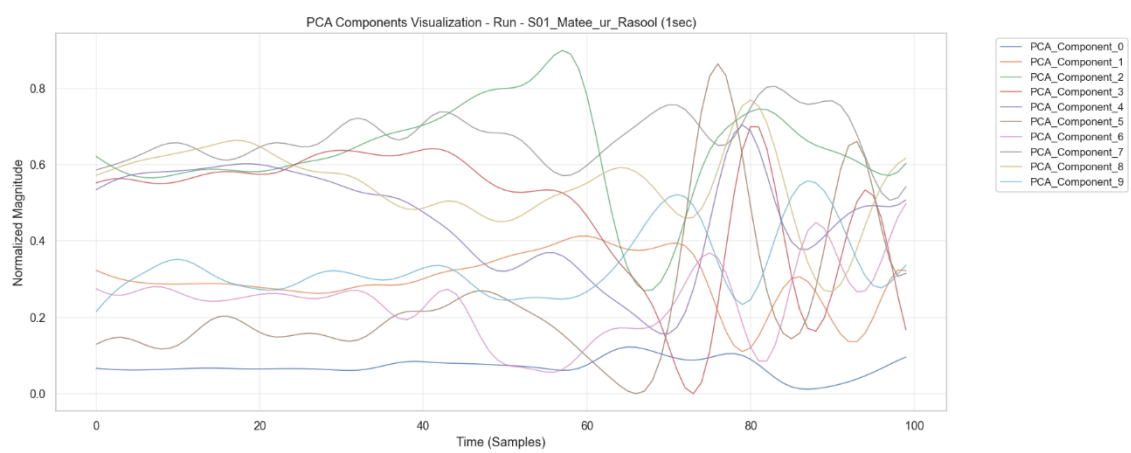


Figure 3:Processed Data

3. Sit/stand

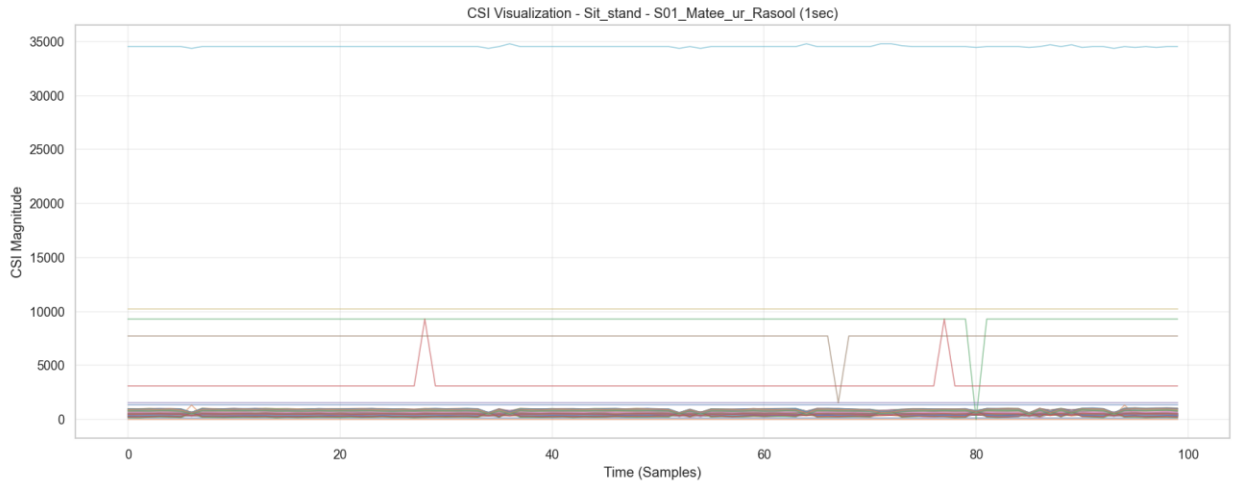


Figure 6:Raw Data for sit/stand(S01)

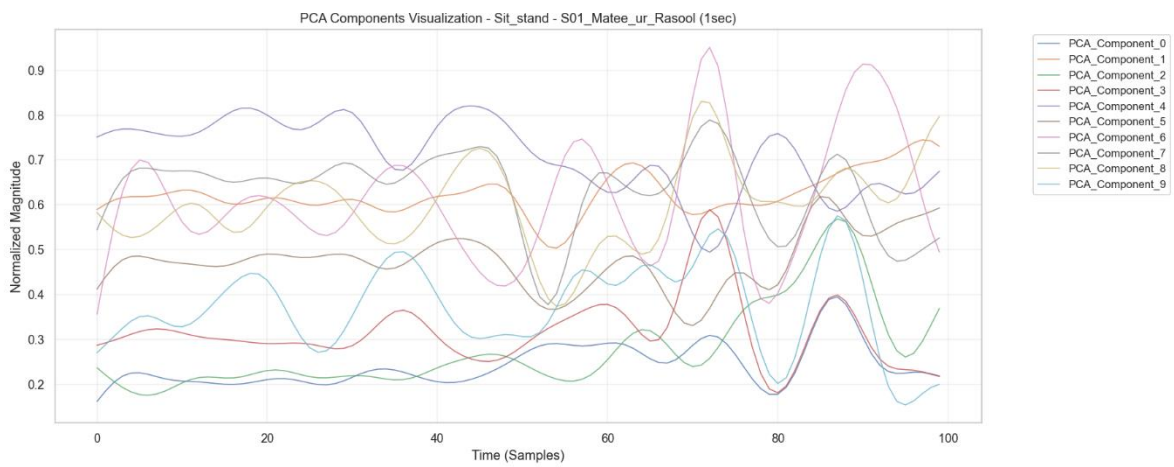


Figure 5: Processed data

4. Jump

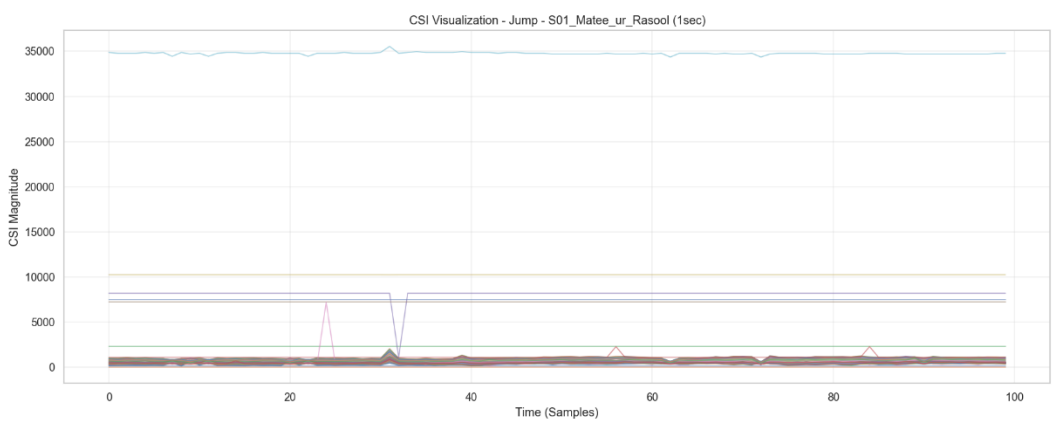


Figure 7:Raw Data for Jump(S01)

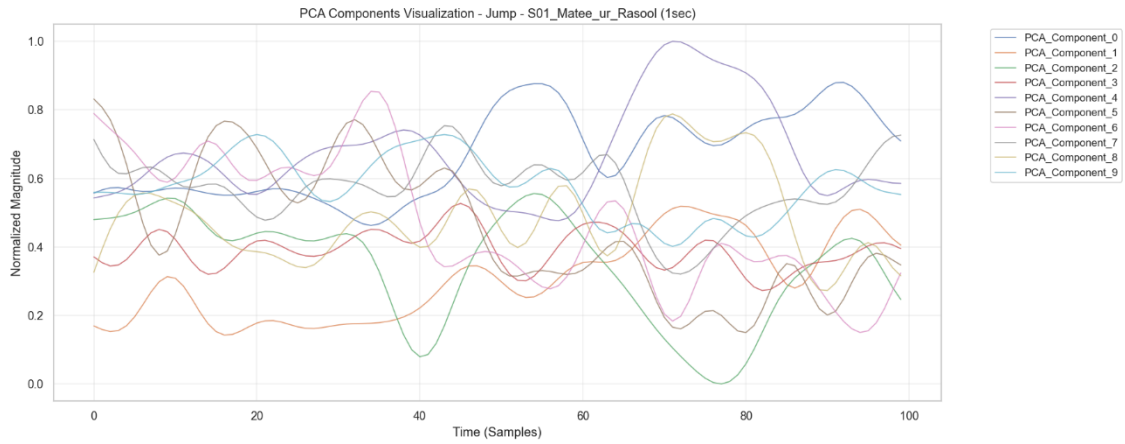


Figure 8:Processed data(S01)

5. Fall

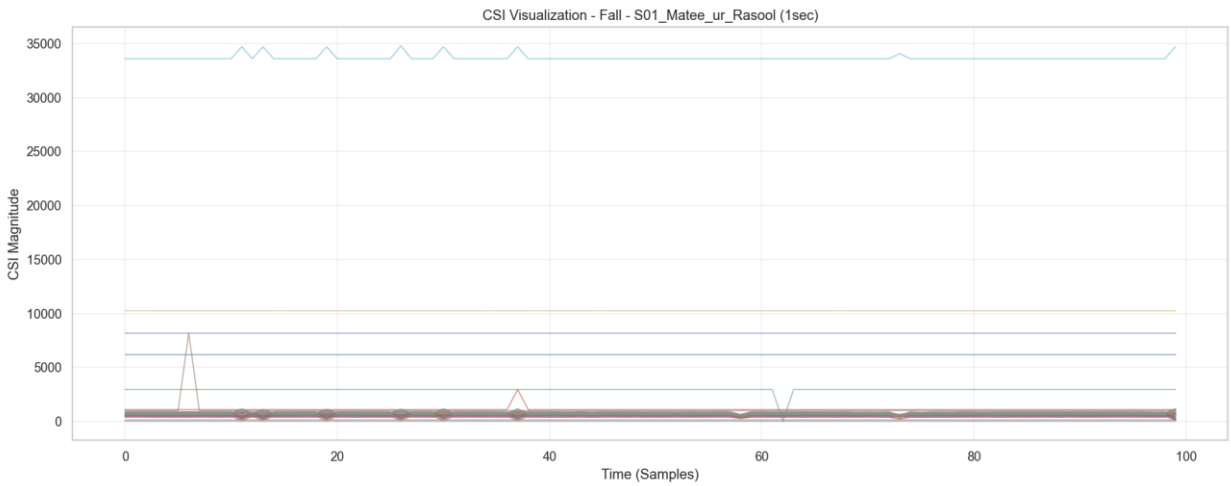


Figure 9:Raw Data Fall(S01)

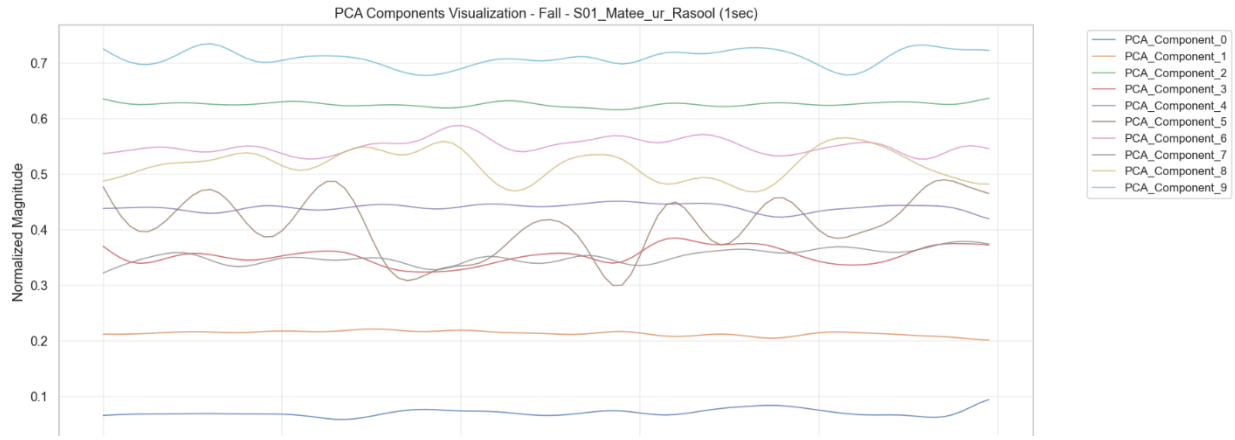
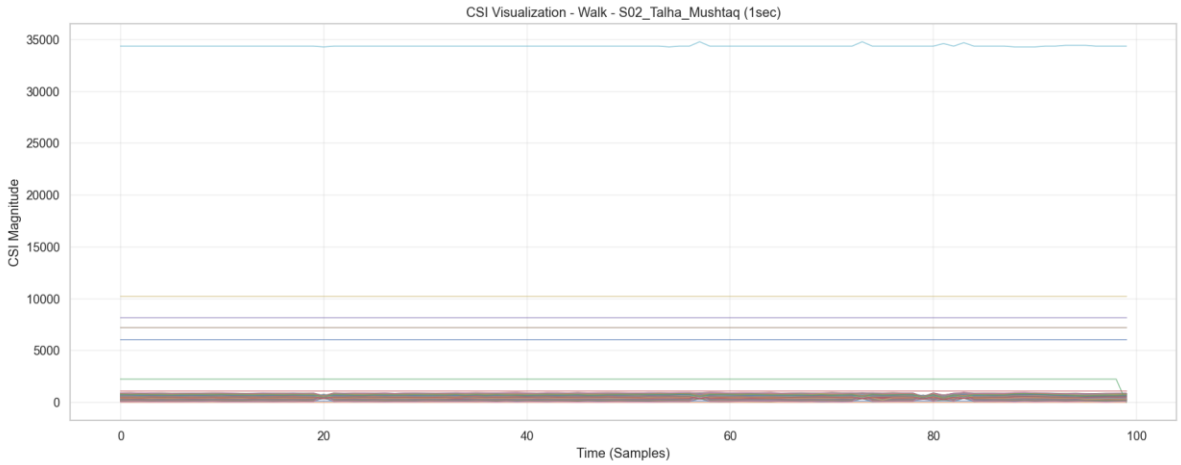
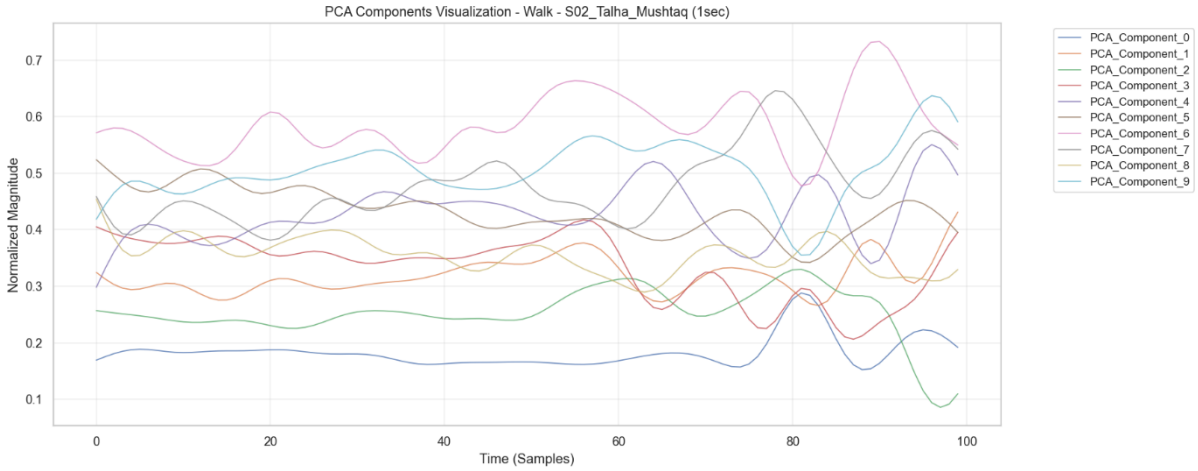


Figure 10:Processed Data

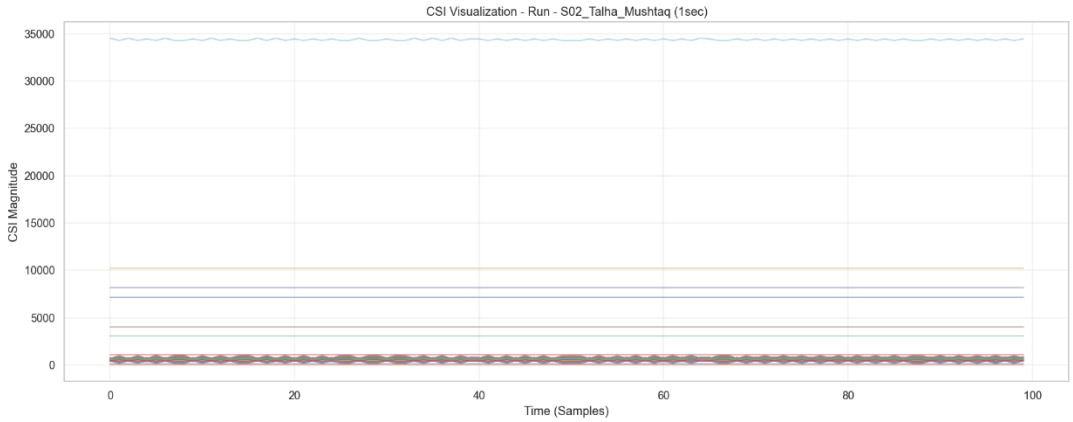
2. Subject 2

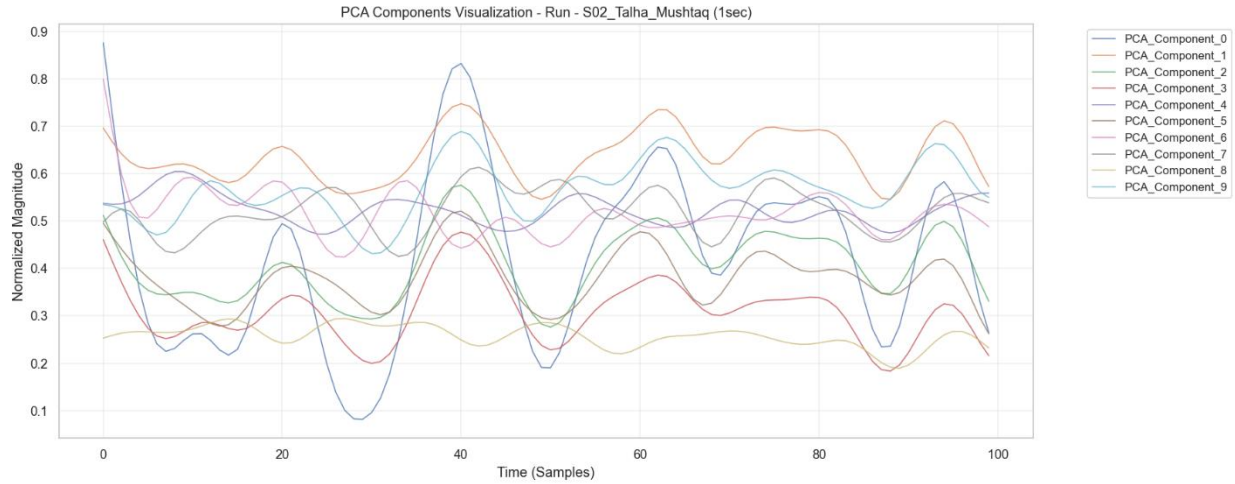
1. Walk



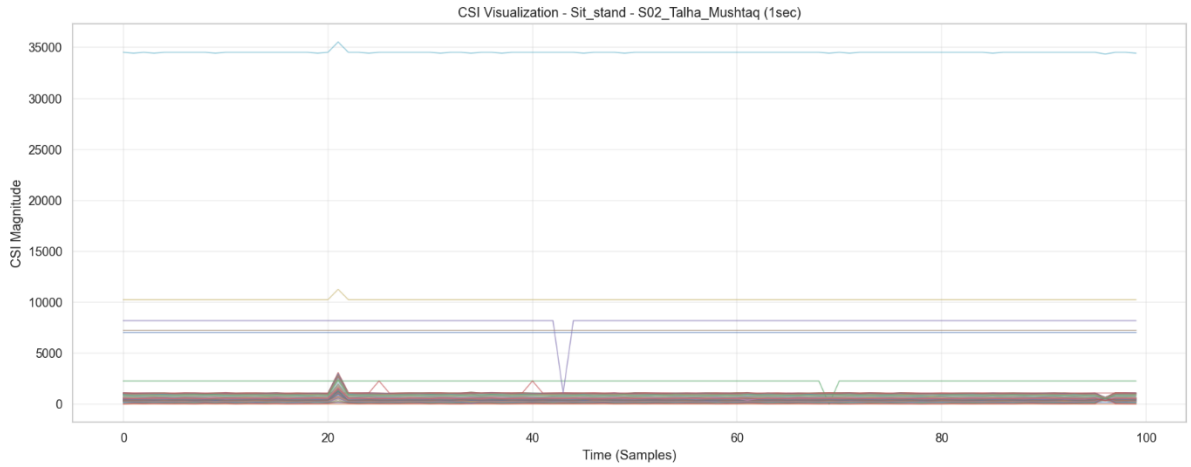


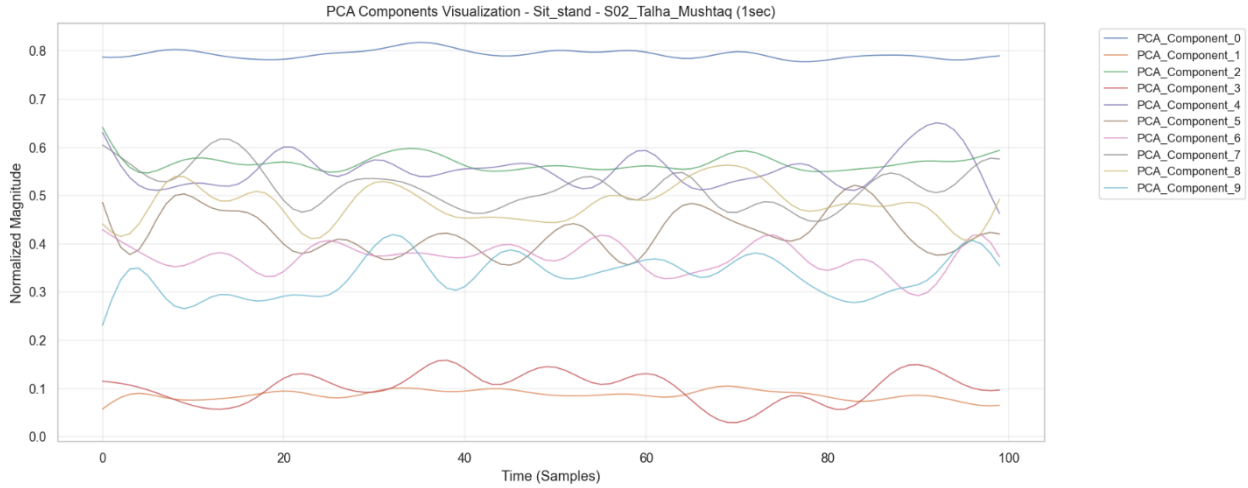
2. Run



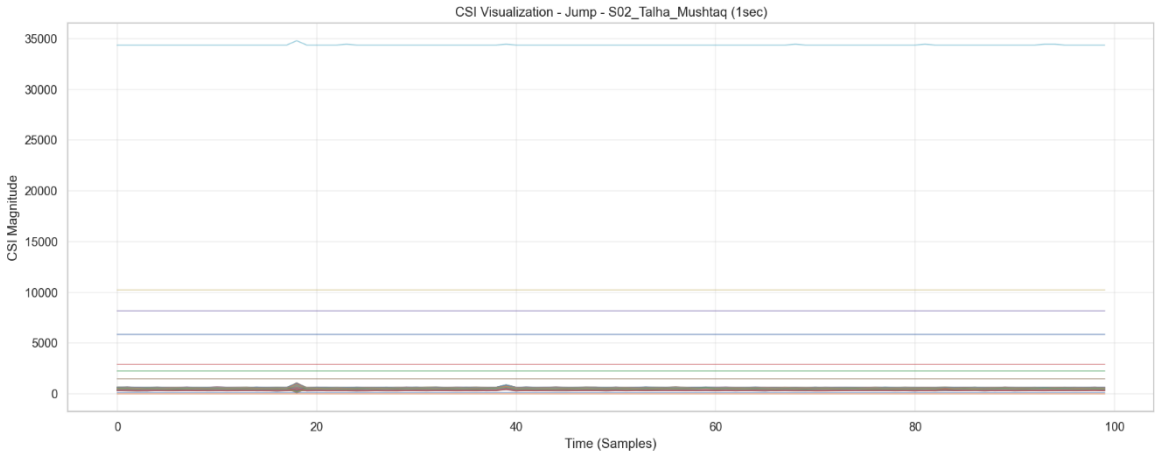

PCA Components Visualization - Run - S02_Talha_Mushtaq (1sec)
Normalized Magnitude
Time (Samples)
PCA_Component_0
PCA_Component_1
PCA_Component_2
PCA_Component_3
PCA_Component_4
PCA_Component_5
PCA_Component_6
PCA_Component_7
PCA_Component_8
PCA_Component_9

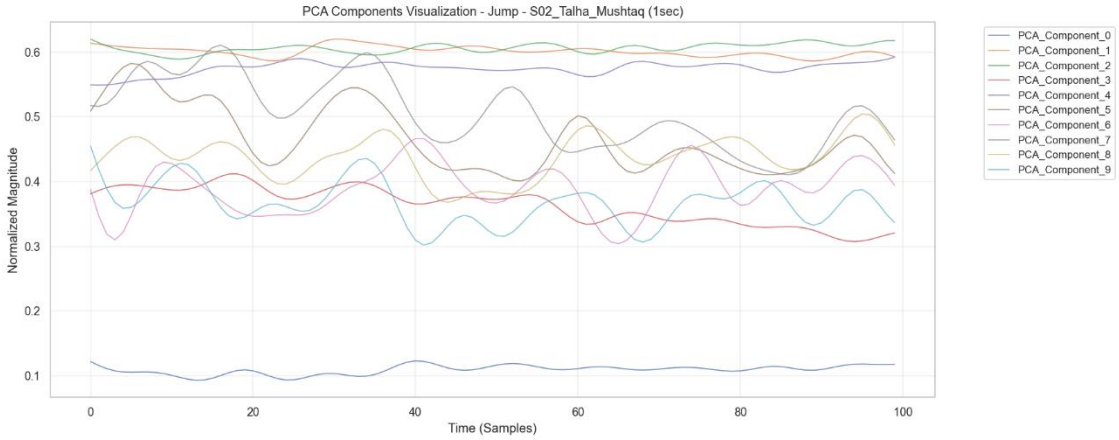
3. Sit/Stand


CSI Visualization - Sit_stand - S02_Talha_Mushtaq (1sec)
CSI Magnitude
Time (Samples)

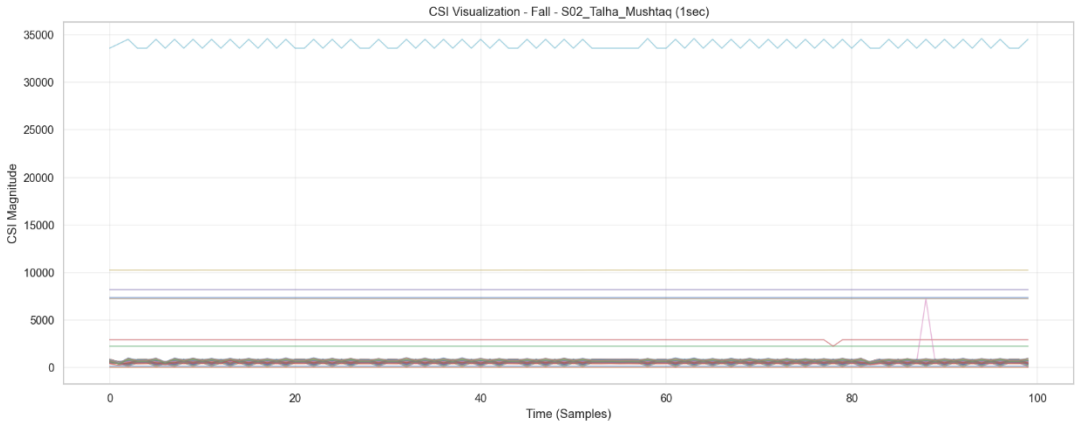

PCA Components Visualization - Sit_stand - S02_Talha_Mushtaq (1sec)
Normalized Magnitude
Time (Samples)
PCA_Component_0
PCA_Component_1
PCA_Component_2
PCA_Component_3
PCA_Component_4
PCA_Component_5
PCA_Component_6
PCA_Component_7
PCA_Component_8
PCA_Component_9

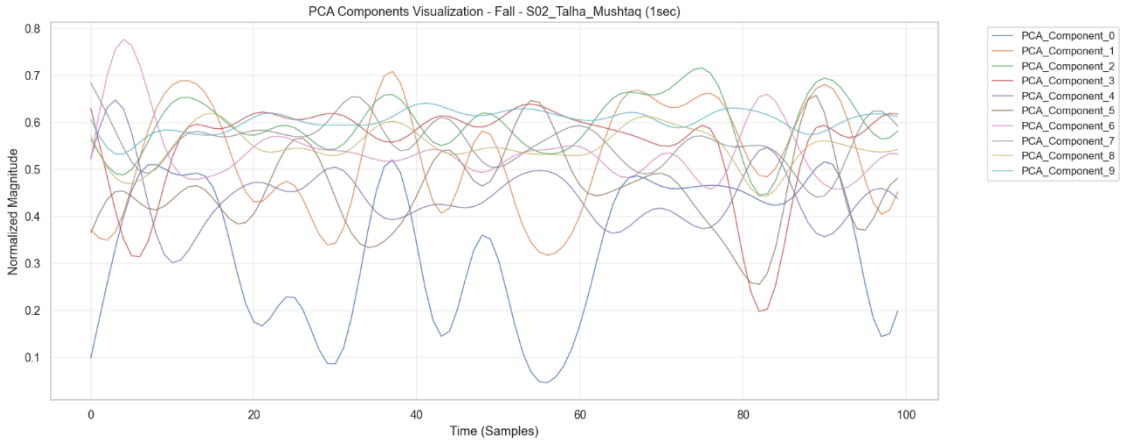
4. Jump


CSI Visualization - Jump - S02_Talha_Mushtaq (1sec)
CSI Magnitude
Time (Samples)


PCA Components Visualization - Jump - S02_Talha_Mushtaq (1sec)
Normalized Magnitude
Time (Samples)
PCA_Component_0
PCA_Component_1
PCA_Component_2
PCA_Component_3
PCA_Component_4
PCA_Component_5
PCA_Component_6
PCA_Component_7
PCA_Component_8
PCA_Component_9

5. Fall


CSI Visualization - Fall - S02_Talha_Mushtaq (1sec)
CSI Magnitude
Time (Samples)


PCA Components Visualization - Fall - S02_Talha_Mushtaq (1sec)
Normalized Magnitude
Time (Samples)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
PCA_Component_0
PCA_Component_1
PCA_Component_2
PCA_Component_3
PCA_Component_4
PCA_Component_5
PCA_Component_6
PCA_Component_7
PCA_Component_8
PCA_Component_9